

TMDB Movie Data Analysis Report

Rankings, Franchise Performance, and Summary Statistics

Summary

- Extracted and cleaned TMDB (The Movie Database) movie data.
- Generated rankings across revenue, budget, profit, return on investment (ROI), audience votes, ratings, and popularity.
- Compiled franchise-level and director-level performance summaries.

Top Rankings

Figures below are in millions of USD (budget, revenue, profit) unless otherwise noted.

Highest Revenue

Title	Release Date	Budget (\$M)	Revenue (\$M)	Rating	Runtime (min)
Avatar	2009-12-16	237.0	2,923.7	7.61	162
Avengers: Endgame	2019-04-24	356.0	2,799.4	8.24	181
Titanic	1997-12-18	200.0	2,264.2	7.90	194
Star Wars: The Force Awakens	2015-12-15	245.0	2,068.2	7.25	136
Avengers: Infinity War	2018-04-25	300.0	2,052.4	8.24	149

Highest Budget

Title	Release Date	Budget (\$M)	Revenue (\$M)	Rating	Runtime (min)
Avengers: Endgame	2019-04-24	356.0	2,799.4	8.24	181
Avengers: Infinity War	2018-04-25	300.0	2,052.4	8.24	149
Star Wars: The Last Jedi	2017-12-13	300.0	1,334.4	6.75	152
The Lion King	2019-07-12	260.0	1,662.0	7.10	118
Star Wars: The Force Awakens	2015-12-15	245.0	2,068.2	7.25	136

Highest Profit

Title	Release Date	Budget (\$M)	Revenue (\$M)	Profit (\$M)	Rating
Avatar	2009-12-16	237.0	2,923.7	2,686.7	7.61
Avengers: Endgame	2019-04-24	356.0	2,799.4	2,443.4	8.24
Titanic	1997-12-18	200.0	2,264.2	2,064.2	7.90
Star Wars: The Force Awakens	2015-12-15	245.0	2,068.2	1,823.2	7.25
Avengers: Infinity War	2018-04-25	300.0	2,052.4	1,752.4	8.24

Lowest Profit

Title	Release Date	Budget (\$M)	Revenue (\$M)	Profit (\$M)	Rating
Star Wars: The Last Jedi	2017-12-13	300.0	1,334.4	1,034.4	6.75
Incredibles 2	2018-06-14	200.0	1,243.2	1,043.2	7.45
Beauty and the Beast	2017-03-16	160.0	1,266.1	1,106.1	6.97
Frozen	2013-11-20	150.0	1,274.2	1,124.2	7.25
Jurassic World: Fallen Kingdom	2018-06-06	170.0	1,310.5	1,140.5	6.53

Highest ROI

ROI = revenue / budget

Title	Release Date	Budget (\$M)	Revenue (\$M)	ROI (x)	Rating
Avatar	2009-12-16	237.0	2,923.7	12.34	7.61
Titanic	1997-12-18	200.0	2,264.2	11.32	7.90
Jurassic World	2015-06-06	150.0	1,671.5	11.14	6.70
Harry Potter and the Deathly Hallows: Part 2	2011-07-12	125.0	1,341.5	10.73	8.08
Frozen II	2019-11-20	150.0	1,453.7	9.69	7.23

Lowest ROI

ROI = revenue / budget

Title	Release Date	Budget (\$M)	Revenue (\$M)	ROI (x)	Rating
Star Wars: The Last Jedi	2017-12-13	300.0	1,334.4	4.45	6.75
Avengers: Age of Ultron	2015-04-22	235.0	1,405.4	5.98	7.28
Incredibles 2	2018-06-14	200.0	1,243.2	6.22	7.45
The Lion King	2019-07-12	260.0	1,662.0	6.39	7.10
Black Panther	2018-02-13	200.0	1,349.9	6.75	7.36

Most Voted

Title	Release Date	Vote Count	Rating	Budget (\$M)	Revenue (\$M)
The Avengers	2012-04-25	38,845	8.06	220.0	1,518.8
Avatar	2009-12-16	34,265	7.61	237.0	2,923.7
Avengers: Infinity War	2018-04-25	32,340	8.24	300.0	2,052.4
Avengers: Endgame	2019-04-24	28,086	8.24	356.0	2,799.4
Titanic	1997-12-18	27,385	7.90	200.0	2,264.2

Highest Rated

Title	Rating	Vote Count
Avengers: Endgame	8.24	28,086
Avengers: Infinity War	8.24	32,340

Title	Rating	Vote Count
Harry Potter and the Deathly Hallows: Part 2	8.08	22,158
The Avengers	8.06	38,845
Titanic	7.90	27,385

Lowest Rated

Title	Rating	Vote Count
Jurassic World: Fallen Kingdom	6.53	12,866
Jurassic World	6.70	21,711
Star Wars: The Last Jedi	6.75	16,460
Beauty and the Beast	6.97	16,076
The Lion King	7.10	10,808

Most Popular

Popularity is TMDB's internal popularity score.

Title	Popularity Score
Avengers: Infinity War	50.70
The Avengers	46.26
Titanic	36.46
Avengers: Endgame	35.87
Avatar	33.12

Franchise Summary

Performance aggregated by film collection/franchise.

Collection	Movies	Total Budget (\$M)	Mean Budget (\$M)	Total Revenue (\$M)	Mean Revenue (\$M)	Mean Rating
Harry Potter Collection	1	125.0	125.00	1,341.5	1,341.5	8.080
The Avengers Collection	4	1,111.0	277.75	7,776.1	1,944.0	7.953
Avatar Collection	1	237.0	237.00	2,923.7	2,923.7	7.610
The Incredibles Collection	1	200.0	200.00	1,243.2	1,243.2	7.454
Black Panther Collection	1	200.0	200.00	1,349.9	1,349.9	7.364
Frozen Collection	2	300.0	150.00	2,727.9	1,364.0	7.239
The Fast and the Furious Collection	1	190.0	190.00	1,515.4	1,515.4	7.218
The Lion King (Reboot) Collection	1	260.0	260.00	1,662.0	1,662.0	7.096
Star Wars Collection	2	545.0	272.50	3,402.6	1,701.3	6.998
Jurassic Park Collection	2	320.0	160.00	2,982.0	1,491.0	6.617

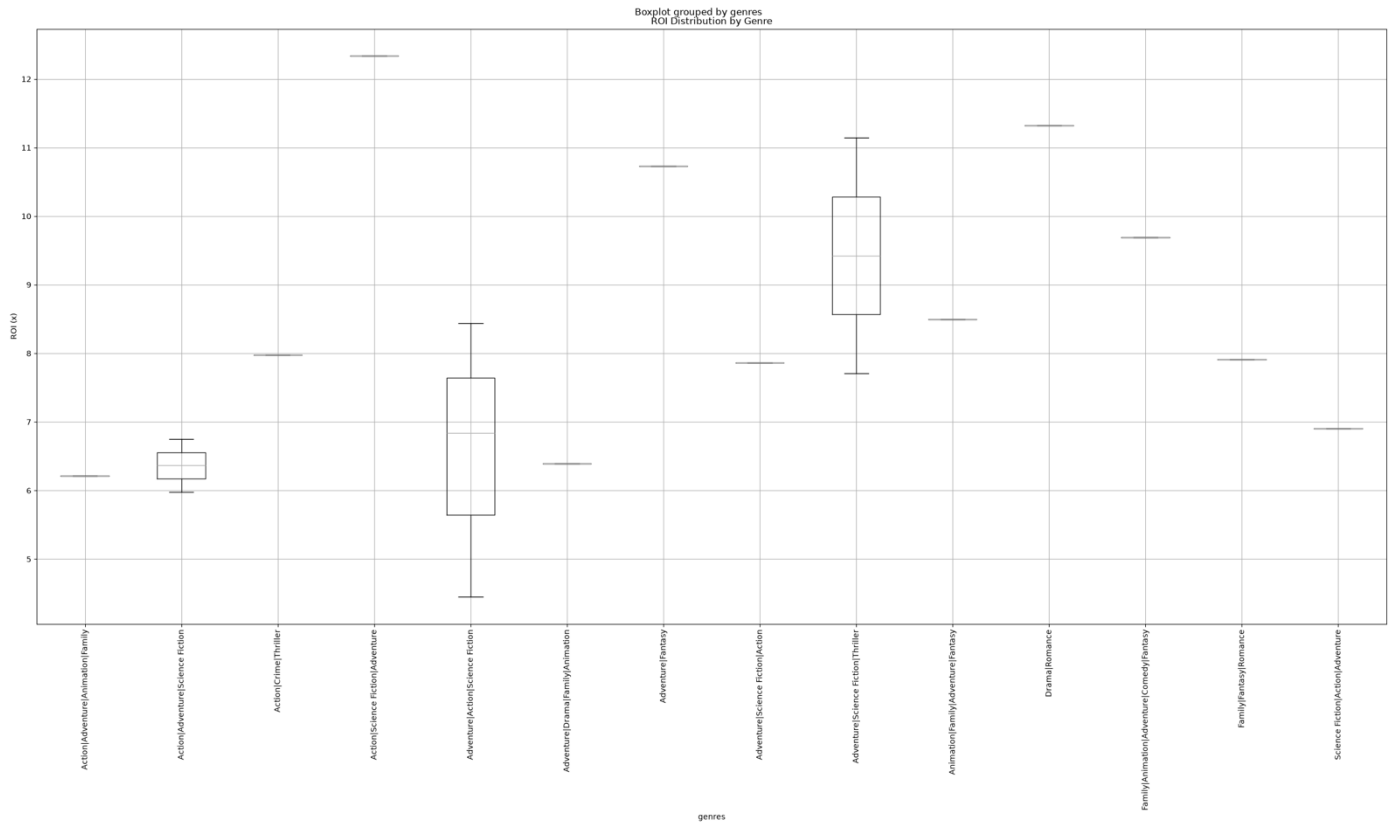
Director Summary

Director data was not captured in the source dataset (all records recorded as missing/NaN). The aggregate below reflects the full 18-movie dataset with director attribution unavailable.

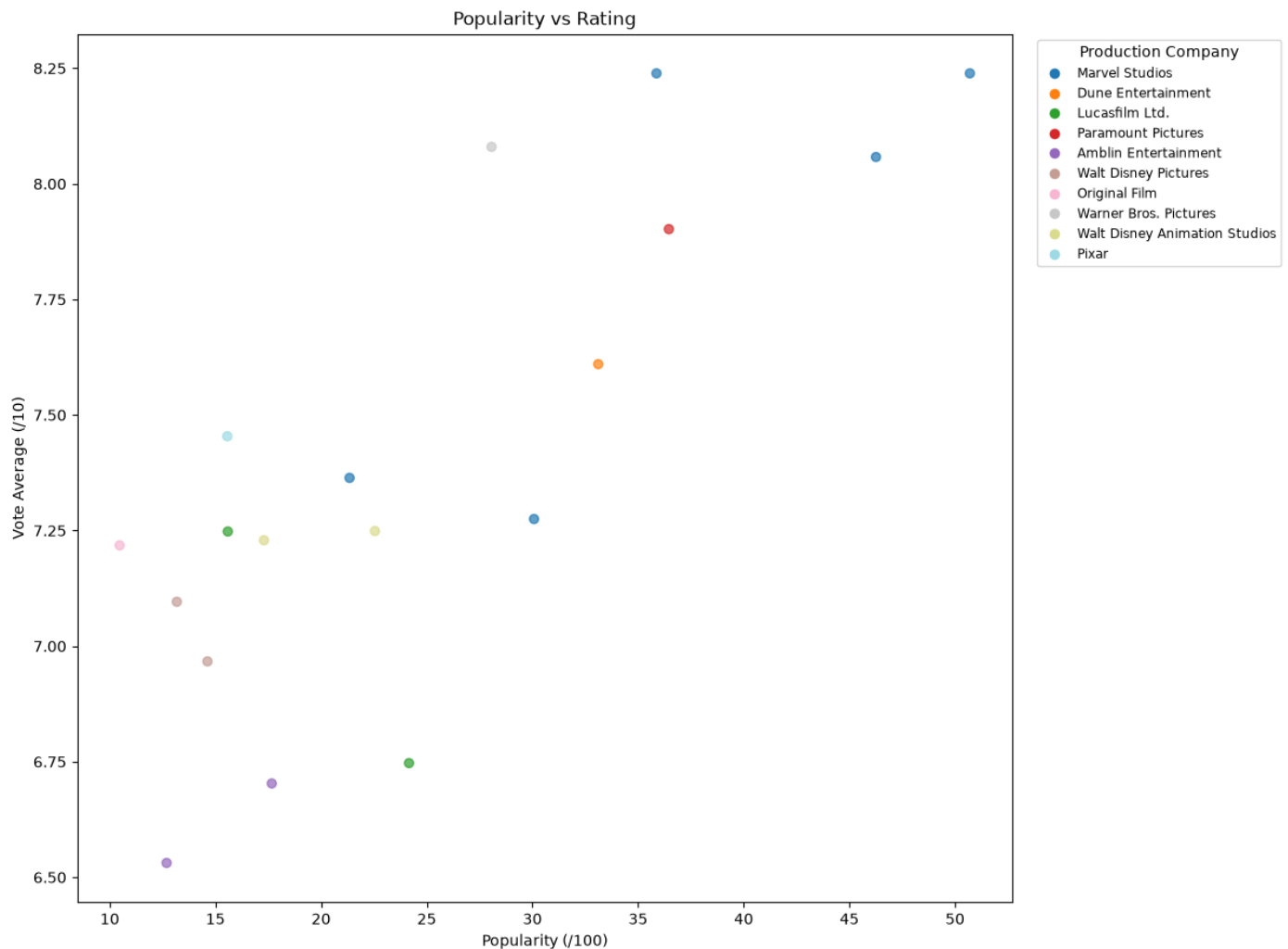
Director	Movie Count	Total Revenue (\$M)	Mean Rating
Not available (missing data)	18	30,454.68	7.401

Visualizations

1. **ROI versus Genres:** varies widely across genre combinations. The ``Action|Science Fiction|Adventure`` combination (**Avatar**) posts the highest ROI (~12.3x), and ``Drama|Romance`` (**Titanic**) follows closely (~11.3x). The ``Adventure|Action|Science Fiction`` genre set — spanning several Marvel and Star Wars titles — shows the widest spread (4.4x–8.4x), reflecting inconsistent returns even within a similar genre mix.



1. **Popularity and audience rating** are not strongly related. **Avengers: Endgame** and **Avengers: Infinity War** (Marvel Studios) share the highest rating (8.24) but sit at very different popularity levels (35.9 vs 50.7), and **Harry Potter and the Deathly Hallows: Part 2** (Warner Bros. Pictures) reaches a similarly high rating (8.08) with only moderate popularity (28.1). At the other end, the *Jurassic World* films (Amblin Entertainment) combine moderate-to-high popularity with the lowest ratings in the set (6.53–6.75), suggesting strong marketing reach didn't translate into critical reception for that franchise.



2. **Revenue vs Budget** — Budget and revenue are loosely correlated but far from linear. **Avatar** ([Dune Entertainment](#)) and **Titanic** ([Paramount Pictures](#)) stand out as the strongest outliers — both earned well over 10x their budget, well above the pack. Marvel Studios (5 films) clusters consistently in the mid-to-high budget/revenue range, showing more predictable returns than the other studios, which each appear only once or twice.

